

Lecture 9

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Group

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CPU Scheduling

- CPU needs to spend time loading files, programs and other things then it will switch over the the I/O and go back and forth from there
- **CPU Burst** - the initial work done by the CPU to set up a process
- Selects from among the processes in memory that are ready to execute and allocates the CPU to one of them
- Non-preemptive/cooperative
 - CPU starts a process and keep it until finished
- Preemptive
 - CPU starts a process and can be interrupted

Dispatcher

- Gives control of the CPU to the process selected by the short-term scheduler
 - switching context
 - switching to user mode
 - jumping to proper location in the users memory

Scheduling Criteria

- **CPU utilization** - keep the same CPU busy (**Maximize**)
- **Throughput** - num. of processes that complete their execution per time unit (**Maximize**)
- **Turnaround time** - amount of time passed to finish execution of a particular process (**maximize**)
- **Waiting time** - total amount of time a process has been waiting in the ready queue (**minimize**)
- **Response time** - amount of time it takes from when a request was submitted until the first response is produced, not output (**minimize**)

Scheduling Metrics

- **Arrival time** - Time the process became ready again
- **Wait time** - Time sent waiting on CPU
- **Service time** - Time spent executing in the CPU
- **Turnaround time** - Total time spent waiting and executing
 - minimizing waiting time will minimize the total turnaround time
 - Service time is constant for the program so target the wait
- Slide 10

First Come First Serve FCFS Scheduling

- processes are assigned the CPU in the order that they request it
- when the running process blocks, the first ready is run next
- when a process gets ready, it is put at the end of the queue
- issue with fairness so using the burst time to help decide who goes first is a good option

Shortest Job First SJF Scheduling

- Associate with each process the length of its next CPU burst. Use the lengths to schedule the process with the shortest time
- Two schemes:
 - Preemptive - assumes the run times are known in advance
 - choose the process whose remaining run time is shortest
 - allows new short jobs to get good service
 - when the burst is the same it will prioritize the newer process
 - Slide 26
 - Non-Preemptive - assumes the run times are known in advance
 - among several equally important ready jobs (or CPU bursts) the scheduler picks the one that will finish the earliest
 - Cannot interfere once the process starts running

Class Question (Non-Preemptive)

- P1 runs first and has a time of 7
- while it is running P2, P3, P4 all arrive
- P3 will run next as it has a burst time of 1
- P2 after that with its time of 4 and an arrival time of 2.0
- Finally P4 will run with its time of 4
 - P1 = 0

- $P2 = (8 - 2) = 6$
- $P3 = (7 - 4) = 3$
- $P4 = (12 - 5) = 7$
- need to subtract the time it started from when it arrived in the queue

Class Question (Preemptive)

- P1 runs first and has a time of 7
- while it is running P2, P3, P4 all arrive
- P3 will run next as it has a burst time of 1
- P2 after that with its time of 4 and an arrival time of 2.0
- Finally P4 will run with its time of 4